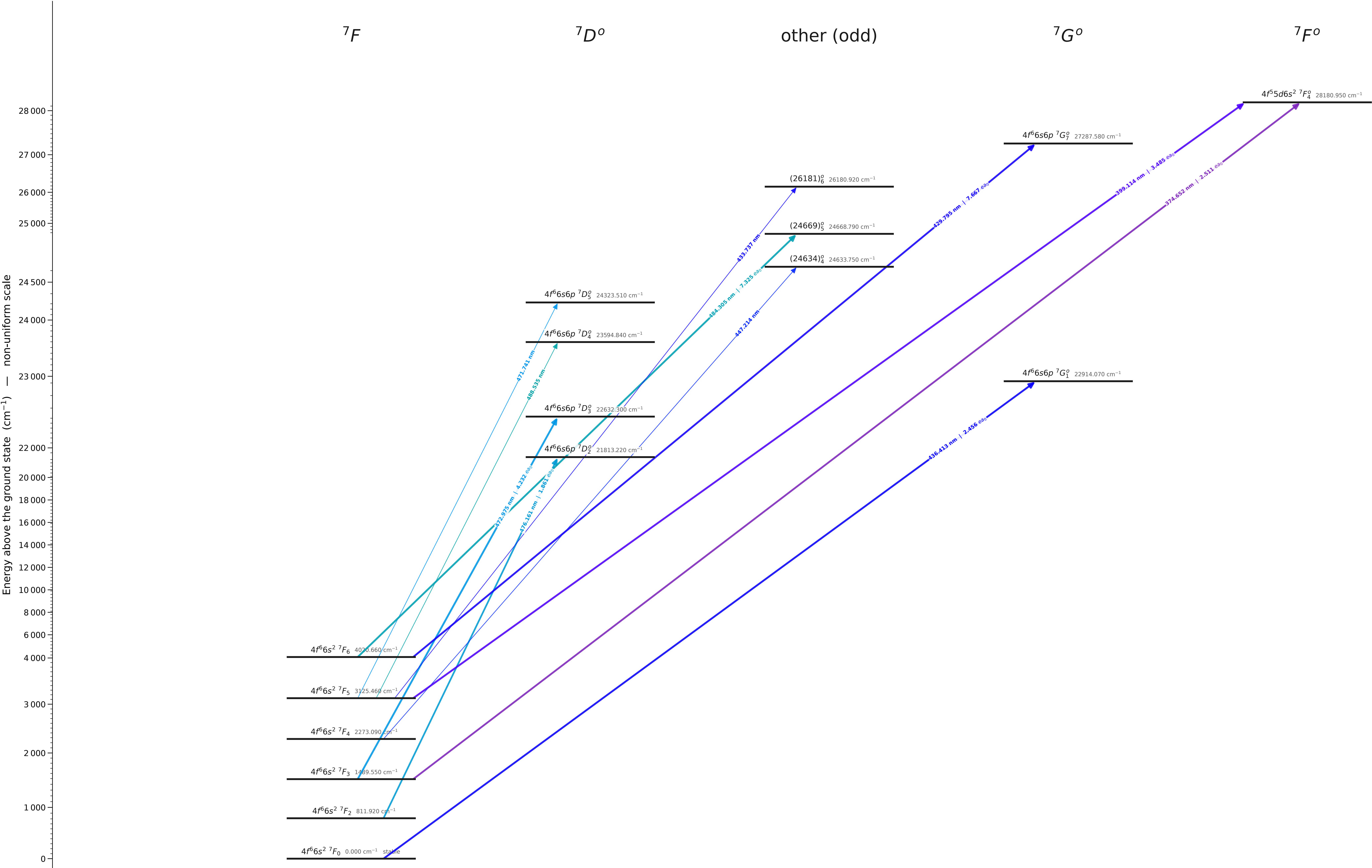
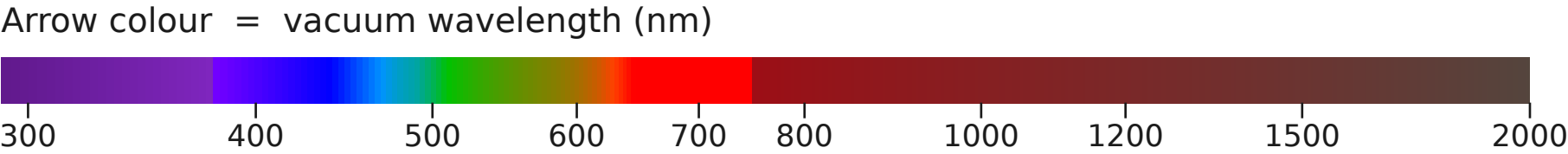


Samarium

Sm (Sm I). Strongest classified transitions below 2 μm (NIST): 16 levels, 11 lines.

Sm⁺ ionisation limit 45519.69 cm⁻¹ = 5.64372 eV (219.685 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e r J' \rangle $ ($e a_0$)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4f^6 6s^2 \ ^7F_3 - 4f^5 5d 6s^2 \ ^7F_4^o$	374.6525	374.5460	800.1880	2.511	2.700e+07	-	-	-
$4f^6 6s^2 \ ^7F_5 - 4f^5 5d 6s^2 \ ^7F_4^o$	399.1141	399.0013	751.1447	3.485	4.300e+07	-	-	-
$4f^6 6s^2 \ ^7F_6 - 4f^6 6s 6p \ ^7G_7^o$	429.7947	429.6739	697.5247	7.667	1.000e+08	-	-	-
$4f^6 6s^2 \ ^7F_0 - 4f^6 6s 6p \ ^7G_1^o$	436.4131	436.2905	686.9465	2.456	4.900e+07	-	-	-
$4f^6 6s^2 \ ^7F_3 - 4f^6 6s 6p \ ^7D_3^o$	472.9754	472.8431	633.8437	4.232	4.900e+07	-	-	-
$4f^6 6s^2 \ ^7F_2 - 4f^6 6s 6p \ ^7D_2^o$	476.1610	476.0279	629.6031	1.861	1.300e+07	-	-	-
$4f^6 6s^2 \ ^7F_6 - (24669)_5^o$	484.3054	484.1701	619.0154	7.325	8.700e+07	-	-	-

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.